Machine Learning for Education Laboratory
ML4ED

Decoding Learning Behaviors: Interpretable Profiles and Their Impact on Academic Outcomes

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DATASET+ RESEARCH QUESTIONS

Dataset: **Education** domain, platform interaction data: Collected from the GoGymi online prep platform for the Swiss Gymnasium entrance exam.

RQ1: Can we identify **interpretable behavioral profiles of students** that are meaningfully related to academic performance?

RQ2: Which behavioral features and dimensions **most strongly drive academic performance**, and how can we explain these relationships interpretably?

RQ1: Learning Behavior Clustering

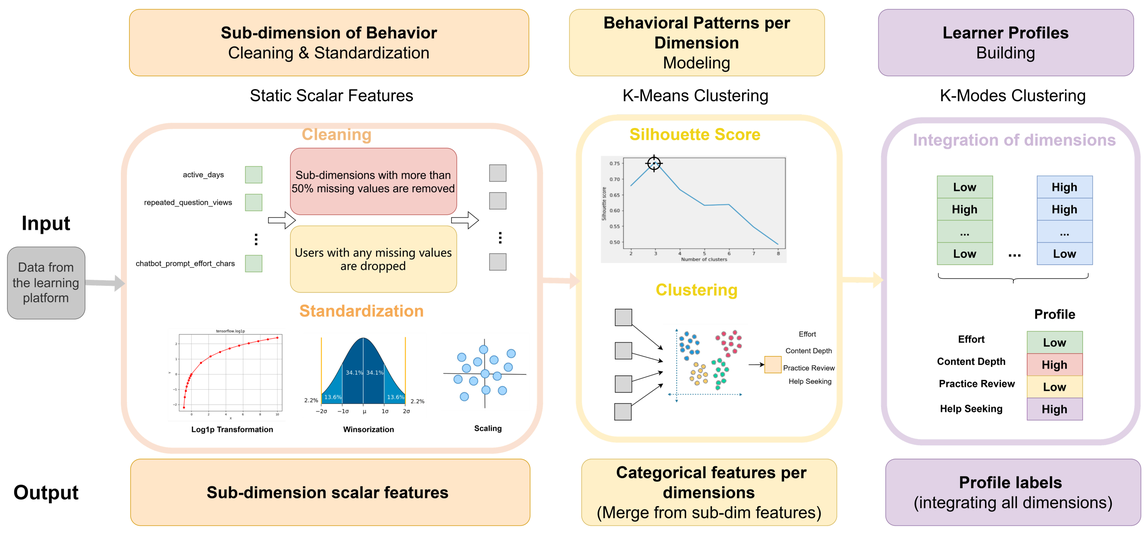


Figure 1: Architecture for Building Student Profiles: Platform data is cleaned and standardized into features, each categorized into a corresponding dimension. The Silhouette Score selects the optimal K for K-Means Clustering on each dimension, assigning every student a cluster label per dimension. These labels are then combined to construct each student's overall profile.

Profile	N	Mean ± Std Score	Rank	Effort	Content Depth	Practice Review	Help Seeking
4	55	0.700±0.14	#1	High Effort	Deep Engagers	Deep Reviewers	Deliberate Askers
5	40	0.690±0.14	#2	Low Effort	Deep Engagers	Deep Reviewers	Deliberate Askers
2	74	0.678±0.15	#3	High Effort	Deep Engagers	Deep Reviewers	Passive Chatters
7	156	0.656±0.14	#4	Low Effort	Deep Engagers	Deep Reviewers	Passive Chatters
0	103	0.630±0.16	#5	Low Effort	Surface Browsers	Deep Reviewers	Passive Chatters
1	31	0.611±0.20	#6	Low Effort	Deep Engagers	Light Reviewers	Deliberate Askers
6	40	0.591±0.18	#7	Low Effort	Deep Engagers	Light Reviewers	Passive Chatters
3	32	0.573±0.22	#8	Low Effort	Surface Browsers	Light Reviewers	Passive Chatters

Table 1: Profile Score Summary (sorted by mean score ratio): The top four profiles *score higher* and *show lower variance* — strong habits produce outcomes that are both better and more predictable. Two patterns stand out: (1) *how* students study matters more than *how long* — Profile 5 outranks Profile 2 despite less platform time; (2) Deliberate Askers (Profiles 4, 5) *consistently beat* Passive Chatters (Profiles 2, 7), even when every other habit is identical.

RQ2: Explainability

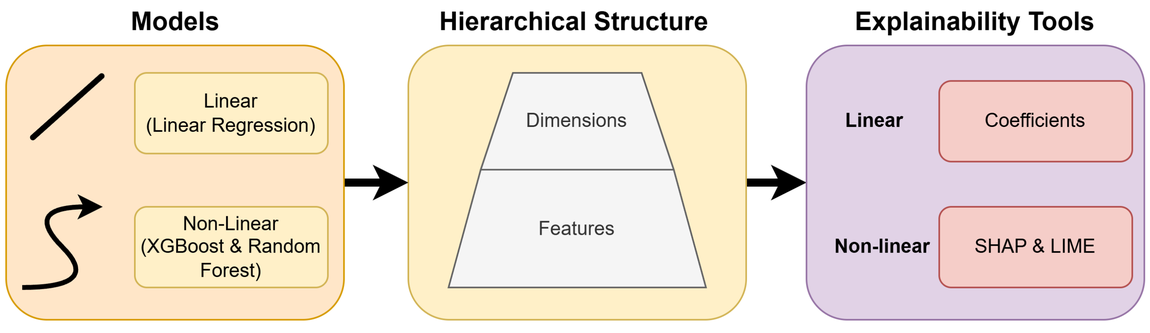


Figure 2: Two-Phase RQ2 Pipeline: Phase 1 runs hierarchical linear regression — Level 1 ranks the four dimensions, Level 2 surfaces the driving features within them. Phase 2 fits XGBoost (primary) and Random Forest (robustness check), then explains them with TreeSHAP (global importance and direction) and LIME (local sanity checks), yielding a dimension ranking and the features that most strongly drive performance.

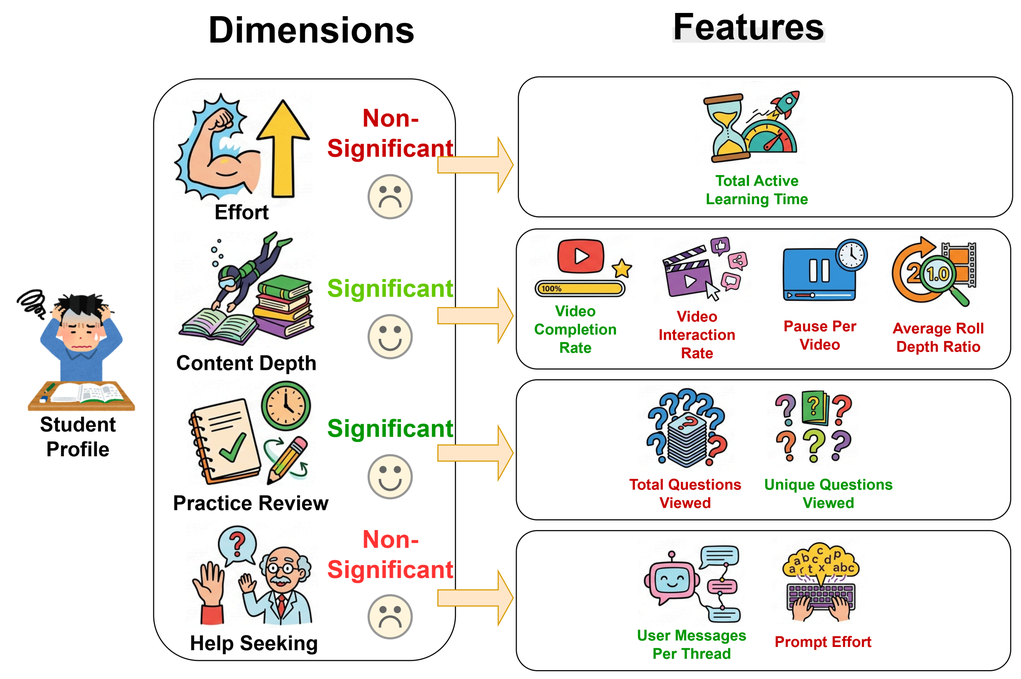


Figure 3: Hierarchical Impact of Learning Behaviors on Academic Performance: We split the analysis into two layers — Dimension and Features. Green indicates variables that are statistically significant for academic performance, while red indicates non-significant ones.

Linear Regression Model

$$\hat{y} = 0.5692 + 0.0263 \cdot \text{Effort} + 0.0355 \cdot \text{ContentDepth} + 0.0555 \cdot \text{PracticeReview} + 0.0140 \cdot \text{HelpSeeking}$$

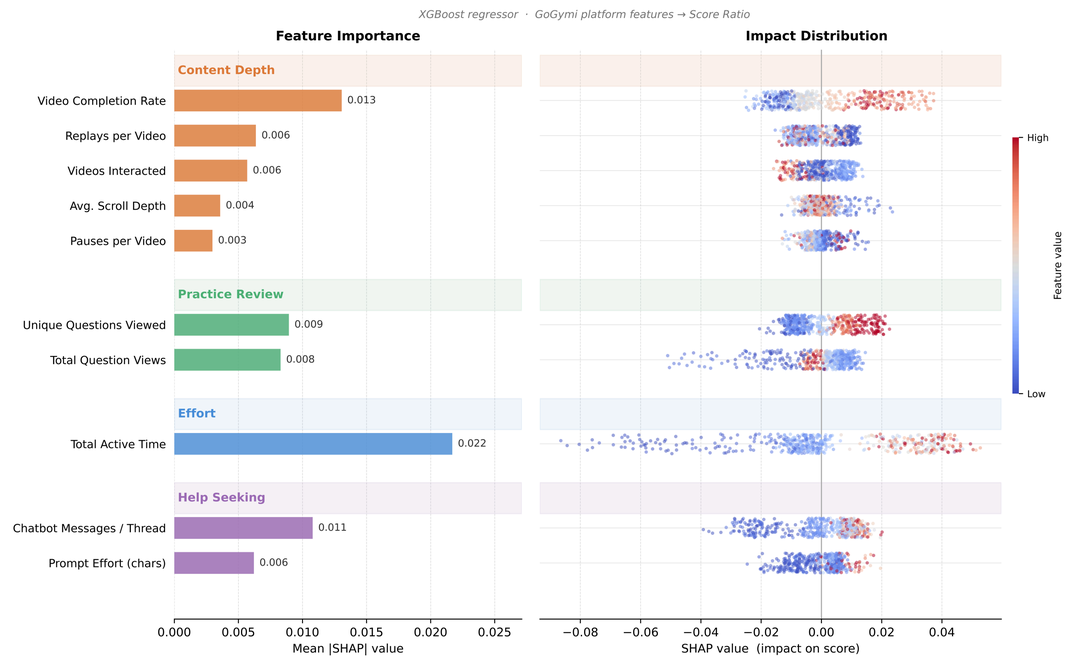


Table 2: Global SHAP: Feature Importance and Direction: Linear and nonlinear models *agree* on the top four drivers in every dimension, lending confidence to the findings. Three patterns stand out: (1) *Total Question Views* and *Videos Interacted* both push scores down — repeatedly revisiting the same content is a red flag; (2) *Chatbot Messages per Thread* pushes scores up — sustained dialogue signals deeper engagement; (3) *Video Completion Rate* (the #2 driver overall) is also positive — *finishing* videos matters more than merely starting them.

CONCLUSION

Result 1

Profile 7 is the highest-leverage target: 156 students (29% of cohort) already engage and review deeply. One nudge toward deliberate help-seeking moves the largest group.

Result 2

All top-ranked profiles share Deep Engagement and Deep Review — Students and parents who equate hours logged with productive study may be optimising the wrong metric.

Takeaway 1

When studying, exploring a wider variety of problems is more beneficial for academic performance than repeatedly reviewing previously seen ones.

Takeaway 2

When interacting with the chatbot, try asking multiple questions and engaging in several rounds of dialogue — this helps you fully understand the problem and resolve your confusion.